Enterprise AI Agentic Framework

A layered architecture for deploying AI agents that operate reliably across clinical, payer, compliance, and legacy boundaries — grounded in DaVita's existing systems and data infrastructure.

The Problem: AI Fails at Domain Boundaries

> AI agents reach **82.8% accuracy** on individual clinical tasks — but drop to **36.3% end-to-end** when steps must chain across domain boundaries. **This is a harness problem, not a model problem.** The framework below is the harness.

Cross-Domain Orchestration Is the Real Moat

Not all harness problems are equal — three categories, three strategies:

HARNESS LAYER	EXAMPLES	MOAT	STRATEGY
Generic Infrastructure Memory, file I/O, state	Task queuing, context windows, retry logic	Low — commodity	Use existing frameworks. Do not build from scratch.
Single-Domain Rules Payer logic, clinical protocols	Prior auth criteria, formulary tiers, ICD-10 coding	**Medium**	Encode as rule engines and domain skills.
Cross-Domain Orchestration Clinical ↔ Payer ↔ Compliance ↔ Legacy	A clinical document's age invalidates a payer auth. A payer decision gates a DME order.	**High — the real moat**	Requires deep operational expertise. Cannot be purchased off-the-shelf.

BOTTOM LINE

The winning architecture is a **domain-aware harness** with cross-domain orchestration. The model handles judgment and edge cases. A co-pilot executes subtask-level work and **escalates at cross-domain decision points** — not an AI left to navigate raw systems alone.

DaVita's existing infrastructure is positioned to deliver exactly this.

The Foundation Is Already Built

The clinical systems, data infrastructure, and AI data services required to power an agentic framework are either active or in delivery. The ask is not to build from scratch — it is to activate the next layer on top.

DAVITA AI DATA SERVICES

RAG Indexing

Unstructured documents — policies, clinical notes, protocols — made semantically searchable

SE Ingestion

Forms, tables, and structured documents extracted and ingested into queryable data pipelines

GraphRAG

Cross-domain knowledge relationships — encoding how clinical, payer, and compliance facts interrelate

DLP — coming

PHI protection boundary ensuring HIPAA-safe data handling across all agent workflows

CORE CLINICAL SYSTEMS

The platforms agents read from, act on, and write back to — the data sources and action surfaces of every workflow.

SYSTEM	ROLE IN DAVITA	VALUE FOR AGENTIC AI
CWOW Clinical Source of Truth	2,800 facilities · 73,000+ patients · encounters, treatment records, orders, labs, vitals — the authoritative clinical record	Agents read lab results and clinical state here to detect triggers (e.g., Hgb drop), validate criteria, and write back authorized care plan changes via the Clinical Management Tool
OneView Physician Experience	CPOE, ePrescribe, order signing, and patient review for nephrologists — 38 microservices on GKE	The escalation surface — agents surface cross-domain recommendations and LCPR approval requests directly into physician workflow, enabling one-tap human sign-off without system-switching
Salesforce P360 Patient Engagement	Patient CRM — campaigns, warm welcome, marketing cloud; receives 15+ clinical datasets from CWOW; physician NPI sync from OneView	Agents close the loop — writing care plan updates, triggering patient outreach, and updating risk segmentation after cross-domain workflows complete
EDP Enterprise Data Platform	Bronze → Silver → Gold governance pipeline · 18+ ESSP consumer teams · Dataflow, Dataform, Airflow · GCP-native	The data abstraction layer for all agents — provides clean, governed, versioned clinical data for trend analysis and historical context; AI model results flow back to CWOW through EDP

Enterprise AI Agentic Framework — Four Layers

Knowledge → Skills → Orchestration → Plugins. Each layer does one thing well — and each maps directly to DaVita's active AI Data Services. No new infrastructure required to begin.

Layer 4 — Domain Plugins Turn-key clinical workflows for end users

- **Prior Authorization** — Clinical → Payer → Compliance → Legacy
- **DME Ordering** — Clinical → Payer → Legacy
- **Appeals Management** — Compliance → Payer → Clinical docs
- **Patient Journey Navigator** — Clinical → CRM → Payer transitions

↕

Layer 3 — Cross-Domain Orchestration The real moat

CANNOT BE PURCHASED OFF-THE-SHELF

Encodes how clinical facts affect payer decisions affect compliance gates — the dependencies no single system captures

Clinical ↔ Payer ↔ Compliance ↔ Legacy

Human escalation at cross-domain decision points

↕

Layer 2 — Domain Skills Executable domain expertise — one skill per domain, reusable across all plugins

- **Clinical Assessment** — EHR read · lab interpretation · clinical criteria · eligibility
- **Payer Navigation** — Formulary lookup · auth rules · medical necessity drafting
- **Compliance Verification** — HIPAA · ICD-10 validation · NPI · audit trail
- **Legacy Bridge** — Fax · Epic · older payer portals — no manual re-entry

↕

Layer 1 — Knowledge Foundation Multi-modal knowledge sources — what agents retrieve and reason from

- **Markdown / Agent Knowledge** — PDLC artifacts, meeting notes, discussions, agent-native context `Direct Access`
- **Datawarehouse / Lakehouse** — Structured EHR data, IoT vitals — governed via EDP Bronze→Silver→Gold `EDP Gold ✓`
- **Documents** — SOP, policies, clinical protocols — semantic retrieval `RAG Indexing ✓`
- **Cross-Organization** — Cross-domain relationships — clinical, payer, compliance interdependencies `GraphRAG ✓`

DLP SECURITY BOUNDARY — HIPAA-SAFE · PHI PROTECTION · COMING SOON

Three-Phase Roadmap

The framework deploys in three phases, building on DaVita's existing AI Data Services:

PHASE 1

Foundation

Months 1–3

- Knowledge Foundation — EDP + AI Data Services
- Clinical Skill + Payer Skill (Layer 2)
- Prior Authorization as validation scenario — see Cross-Domain Use Case →
- Human-in-the-loop at decision points

PHASE 2

Cross-Domain Orchestration

Months 4–8

- Layer 3 — clinical ↔ payer ↔ compliance dependencies
- Compliance Skill + Legacy Bridge Skill
- DME Ordering + Appeals Management plugins
- Reduce human escalation rate

PHASE 3

Scale & Governance

Months 9–18

- DLP Security Boundary across all workflows
- Patient Journey Navigator + additional plugins
- RBAC, adoption metrics, audit trails
- Expand to adjacent clinical programs

Prior Authorization — ESA Therapy

The goal: **build and submit a compliant clinical evidence package to secure ESA prior authorization** — gathering patient data, validating CMS protocols, detecting hidden payer rules, and obtaining physician sign-off across four separate domains. Today: 4–7 days, 3–5 people. With the agentic framework: same-day, one physician touch point.

Each step maps to a framework layer — the framework shown in practice on a real DaVita workflow.

PATIENT CASE INTRODUCTION

PATIENT

Maria, 67

Condition: End-stage renal disease (ESRD)
Treatment: Hemodialysis, 3× per week
Need: ESA authorization renewal — expires 14 days
Goal: Submit evidence package to insurance to secure prior authorization for ESA therapy

CLINICAL SYSTEM (CWOW) KNOWS

9.2 → 9.5 → 9.8

Hgb g/dL over 12 weeks · below 10 ✓

· IV iron confirmed (not oral) ✓
· No contraindications ✓
· Clinically, criteria are met
CWOW does not know the insurance rule about trends

INSURANCE SYSTEM (PAYER) HAS

· Rule A: Hgb must be < 10 g/dL ✓
· Rule B: Must be on iron therapy ✓

Hidden Rule C — Improvement Clause:
"If Hgb rises consistently over 3+ months, deny authorization — regardless of absolute value."

This rule is not shared with the clinical system. CWOW cannot see it.

THE GAP

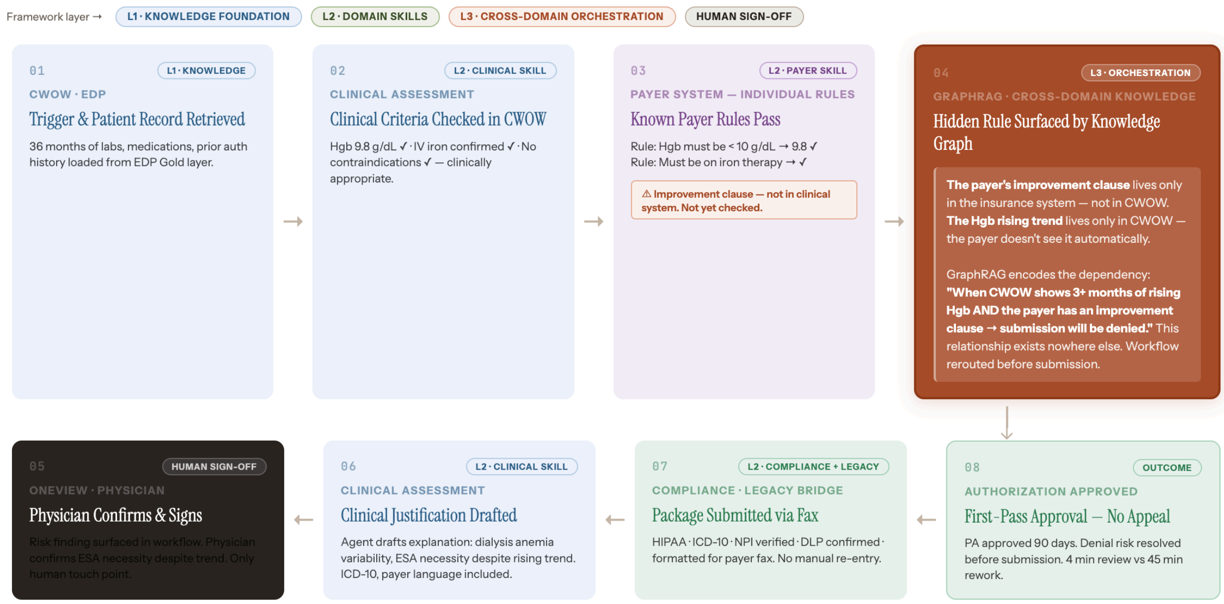
CWOW has Maria's Hgb trend data. The payer has the improvement clause. **Neither system exposes this to the other — and this is not an oversight. It is structural.**

Payer rules are not database fields. They exist as **PDF policy documents, contract language, and prior auth criteria letters** — not queryable APIs. DaVita works with hundreds of insurance companies, each updating their own formularies independently. Building a technical sync would require a custom integration per payer, constant maintenance, and payer cooperation that is rarely available.

The un-synced state is the *inherent architecture of healthcare* — not a gap waiting to be engineered away with an ETL pipeline. This is why **L1 ingests payer policy PDFs via RAG Indexing Service (active)**, and **L3 GraphRAG Ingestion Service (active) encodes the logical relationship** between clinical patterns and payer rules — so when rules change, you update the knowledge graph, not a fragile point-to-point integration.

`Insurance policies build the graph.  Patient data queries it.`

CROSS-DOMAIN WORKFLOW



VALUE PROPOSITION OF THE AGENTIC FRAMEWORK

WITHOUT — WHAT GOES WRONG

- ✗ CWOW check ✓ · Payer Rule A & B ✓ — Rule C (improvement clause) never surfaced
- ✗ Submitted with false confidence → payer detects rising trend → **DENIED**
- ✗ 2–4 week appeal · therapy interrupted · staff hours lost to rework

4–7 days · high denial rate

WITH — HOW L3 CLOSES THE GAP

- ✓ **L3 GraphRAG detects Rule C** — encodes the dependency between CWOW trend and payer policy before submission
- ✓ Workflow rerouted · physician adds justification · L2 Skills compile evidence package
- ✓ First-pass approval · therapy uninterrupted · rule reusable for all future cases

Same day · 1 human touch point · 4 min review